

- Q.27 Explain the concept of view ports and how to change the layout from Model to Paper space.
- Q.28 Explain the different types of robot motions and configurations.
- Q.29 What is the purpose of dimensioning solids in AutoCAD? Explain with any sketch.
- Q.30 Discuss the different terms used in milling and turning like clearance, Retract, Feed plane, and depth of cut.
- Q.31 Explain the different commands used for creating solids like Region, Extrude, and Revolve features.
- Q.32 Describe the steps involved in a CAM operation.
- Q.33 What is a Robot? Explain different robot sensors and programming languages.
- Q.34 Explain the process of creating fillets and chamfers on a solid model.
- Q.35 What are the advantages of CAD?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain in detail the different operations involved in both turning and milling machines?
- Q.37 Describe the process of tool path generation and verification. Include details on how to set up jobs, specify tools, and back plot the operation?
- Q.38 Explain Flexible Manufacturing Systems (FMS), its features, advantages, and limitations?

No. of Printed Pages : 4
Roll No.

181761C/171761C
/062463

6th Sem / Mech., Mechatronics, CNC Subject:- CAD/CAM/CAD CAM & FMS

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is an advantage of CAD?
- Increased product cycle time
 - Reduced automation
 - Automation and CAD/CAM
 - Slower implementation of CAD/CAM
- Q.2 Which of these is not a command used for creating simple surfaces?
- Revolved surface
 - Rules surface
 - 3D surfaces
 - Union
- Q.3 What are the two types of surfaces mentioned in the syllabus?
- Extrude and Revolved
 - Parametric and non-parametric
 - Fillets and chamfers
 - Union and Subtraction
- Q.4 Which of these is a command for the creation of a solid using a revolved feature?
- Revolved
 - Array
 - Mirror
 - Rotate
- Q.5 What is the purpose of back plotting and verification of an operation?

- a) To define and chain the geometry
 - b) To convert the tool path into NC code
 - c) To specify the tools and machining parameters
 - d) To set up jobs and verify operations
- Q.6 What is a key step involved in a CAM operation?
- a) Creation of fillets
 - b) Dimensioning of solids
 - c) Introduction to CAD /CAM
 - d) Post-processing
- Q.7 What is the process of converting the generated tool path to NC code called?
- a) Post-Processing b) Tool path generation
 - c) Setting up the jobs d) Backplotting
- Q.8 Which of the following is a component of a Flexible Manufacturing System (FMS)?
- a) Robot configuration
 - b) Robot motions
 - c) Material handling system
 - d) Robot programming languages
- Q.9 Which of the following is a turning operation?
- a) Pocketing b) Facing
 - c) Contouring d) Circular tool paths
- Q.10 What is the process of setting up parameters for the transfer of programs to a CNC machine called?
- a) Communication b) Post-processing
 - c) Backplotting d) Chaining the geometry

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What are the two types of FMS?
- Q.12 Name any two CAD software.

- Q.13 What are the three commands used for the creation of solids?
- Q.14 What are the two types of flexibility in a manufacturing system?
- Q.15 Explain fillet & Chamfer command
- Q.16 List three application of Robots?
- Q.17 What are the four components of a Flexible Manufacturing System (FMS)?
- Q.18 Explain in brief about Robot Motion
- Q.19 What are two solid editing commands used to modify solid edges and faces?
- Q.20 Name the four tools materials used in CNC operation?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the concept of parametric and non-parametric surfaces.
- Q.22 Describe the steps involved in the creation of a composite solid using Boolean functions like Union, Subtraction, and Intersection.
- Q.23 What are the different types of tools and devices used in CNC?
- Q.24 What is Flexible Manufacturing System (FMS)? Explain the principles of flexibility.
- Q.25 Describe the process of setting up jobs, specifying tools, and backplotting for tool path generation.
- Q.26 Explain :- 3D Array, mirror and Rotate command?